Tensor product

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This write-up is going to be more of a "collection of facts" than "actually stating and proving stuff". We shall see the general definition of the tensor product and see what nice connections we can find.

Consider a multicategory $\boldsymbol{C}$. Given arbitrary objects $A_1, ..., A_k, X \in \mathrm{Obj}(\boldsymbol{C})$, let us name $\boldsymbol{C}_k(A_1, ..., A_k; X)$ as the set of multimorphisms of type $A_1, ..., A_k \to X$.

Given objects A and B in $\boldsymbol{C}$, if there exists a tuple $(T \in \mathrm{Obj}(C), \varphi \in \boldsymbol{C}_2(A, B; T))$ such that given an arbitrary object X, any multimorphism from A, B to X factors uniquely through φ, then (T, φ) is known as the **tensor product** of A and B.

Referring to Claudio Hermada's *Representable multicategories*,

> 8.1. Definition. A multicategory $\mathbb{M}$ is said to be *representable* if
>
> (1) for every tuple of objects $\vec{x} = \langle x_1, ..., x_n \rangle$ there exists a *universal arrow*, $\pi_{\vec{x}}: \langle x_1, ..., x_n \rangle \to \otimes \vec{x}$, in the sense that precomposition with this arrow induces an isomorphism
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> $$\mathbb{M}(\langle x_1, ..., x_n \rangle, y) \cong \mathbb{M}(\langle \otimes \vec{x} \rangle, y)$$
>
> natural in y, and
>
> (2) universal arrows are closed under composition.

> 8.2. *Remark*. The universality condition in (1) above can also be expressed as follows. Recall from Definition 6.7 the underlying category of linear morphisms of a multicategory $\mathbb{M}$, written $\bar{\mathbb{M}}$. For any tuple of objects $\langle x_1, ..., x_n \rangle$, we get a functor $\mathbb{M}(\langle x_1, ..., x_n \rangle, _): \bar{\mathbb{M}} \to \mathscr{S}et$. The existence of a universal arrow for $\vec{x}$ amounts then to the representability of this functor.
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> As for condition (2), it is a necessary requirement to ensure the associativity of the tensor, e.g., $(x \otimes y) \otimes z \cong x \otimes (y \otimes z)$ actually means that both $\pi_{(x \otimes y), z} \circ \langle \pi_{x, y}, id_z \rangle$ and $\pi_{x, y \otimes z} \circ \langle id_x, \pi_{y, z} \rangle$ are universal arrows for $\langle x, y, z \rangle$, cf. Definition 9.6.

Note that the mere existence of all n-ary tensor products is not sufficient to guarantee that the tensor product is associative.

Therefore, in a representable multicategory, all n-ary tensor products exist and binary tensor product is associative.

We can define a multicategory $\boldsymbol{M_1}$ where the objects are from k-Vect, and the set of multimorphisms $\boldsymbol{M_1}(A_1, ..., A_n, B)$ is the set of all multilinear maps from $A_1 \times ... \times A_n$ to B. The set of multimorphisms $\boldsymbol{M_1}(; B)$ is defined as the set of vectors in B itself, and the reason why we do that will be evident soon. Multimorphism composition can be defined easily.

Observe that $\boldsymbol{M_1}$ is a representable multicategory, with the tuple $(A_1, ..., A_n)$ being "represented" by $A_1 \otimes ... \otimes A_n$ and the universal multimorphism from $A_1, ..., A_n$ to $A_1 \otimes ... \otimes A_n$ being the canonical multilinear map.

The reason why we have $\boldsymbol{M_1}(; B)$ as the set of all vectors in B is because we **want** the representing object for the empty list to be $\bigotimes_{A \in \phi} A = k$, where we are thinking of k as a k-vector space. So, we want multilinear maps (multimorphisms in $\boldsymbol{M_1}$) from the empty list to B to be linear maps (morphisms in k-Vect) from k to B, which are in one-to-one correspondence with vectors from B.

We can go a step further and ask why we want $\bigotimes_{A \in \phi}$ to be k. Firstly, any other choice (something not isomorphic to k) for the representing object of the empty list will *not* make $\boldsymbol{M_1}$ into a representable multicategory. Also, as we will see in a later section, k-Vect is a monoidal category (with the monoidal product being $\otimes$), and therefore has a canonical representable category, and the representing object of the empty list is just the unit of the monoidal category, which is k in k -Vect.

We can also define yet another multicategory $\boldsymbol{M_2}$, where the objects are still from k-Vect, but the set of multimorphisms $\boldsymbol{M_2}(A_1, ..., A_n; B)$ is defined as $\prod_{A_i} \mathrm{Hom}_{k\text{-Vect}}(A_i, B)$. Observe that $\boldsymbol{M_2}(; B)$ is therefore defined as $\prod_{A_i \in \phi} \mathrm{Hom}_{k\text{-Vect}}(A_i, B) = \{*\}$, the empty product.

$\boldsymbol{M_2}$ is a representable multicategory, with the $(A_1, ..., A_n)$ being represented by $\bigoplus_i A_i$ and the universal morphism being the tuple of inclusion maps.

Observe that again, multimorphisms from the empty list to B are in one-to-one correspondence with morphisms from the representing object of the empty list $\bigoplus_{A \in \phi} A = \{0\}$ to B. The same discussion above goes through, as k-Vect is also a monoidal category with $\oplus$ as the monoidal product.

So, in $\boldsymbol{M_1}$, we get the tensor product to be $\otimes$, and in $\boldsymbol{M_2}$, we get the tensor product to be $\oplus$. The reason why we prefer $\otimes$ as the tensor product over k-vector spaces instead of $\oplus$ will be discussed in a later section.

Any monoidal category $(\boldsymbol{D}, \otimes, 1)$ canonically defines a canonical representable multicategory where the objects are from $\boldsymbol{D}$, and the set of multimorphisms from $A_1, ..., A_n$ to B is defined as $\mathrm{Hom}_{\boldsymbol{D}}(A_1 \otimes ... \otimes A_n, B)$. The set of multimorphisms from the empty list to B is defined as $\mathrm{Hom}_{\boldsymbol{D}}(1, B)$. The n-tuple $(A_1, ..., A_n)$ has $(A_1 \otimes ... \otimes A_n)$ as its representing object, with the universal multimorphism being just $\mathrm{id}_{A_1 \otimes ... \otimes A_n}$.

Any closed category $\boldsymbol{D}$ canonically defines a (closed) multicategory where the objects are from $\boldsymbol{D}$, and the set of multimorphisms from $A_1, ..., A_n$ to B Try again later.

Try again later. : Now tell adjoint property of closed monoidal categories.

Let us go back to our multicategories $\boldsymbol{M_1}$ and $\boldsymbol{M_2}$.

Observe that $\boldsymbol{M_1}$ is the canonical representable multicategory induced by the monoidal category k-Vect (where the monoidal product is $\otimes$), while $\boldsymbol{M_2}$ is the canonical representable multicategory induced by the monoidal category k-Vect (where the monoidal product is $\oplus$).

Now, k-Vect has the additional structure of being a closed category, with the internal hom $[V, W]$ being the k-vector space structure on $\mathrm{Hom}_k(V, W)$.

The monoidal category k-Vect (with $\otimes$ as the monoidal product) becomes a closed monoidal category under the above interpretation of k-Vect as a closed category. Therefore, here the tensor product in $\boldsymbol{M_1}$ satisfies the nice adjoint property that for any k-vector space B,

$$\mathrm{Hom}_k(A \otimes B, C) \overset{\text{natural in } A,C}{\cong} \mathrm{Hom}_k(A, [B, C]) \Leftrightarrow - \otimes B \dashv [B, -]$$

However, the monoidal category k-Vect (with $\oplus$ as the monoidal product) doesn't form a closed monoidal category under the same interpretation of k-Vect as a closed category, and therefore, we get no such thing.

So, in this particular sense, $\otimes$ is a "nicer" tensor product on k-Vect than $\oplus$, since it uses the fact that k-Vect can be made into a closed monoidal category, and we also get the nice property that the functors $- \otimes B$ and $[B, -]$ are adjoint functors.

References:

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